Cisco PE QoS Congestion Event SOP for Node KOR-PE-01 at Koramangala Datacenter

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_6827

Description

The Cisco PE (Provider Edge) role is susceptible to QoS congestion events, manifesting as microbursts, interface tail drops, or QoS queue starvation, due to the inherent complexities of implementing differentiated services within the MPLS (Multiprotocol Label Switching) framework. The PE device's reliance on the MQC (Modular Quality of Service Command-Line Interface) to configure QoS policies introduces a high degree of variability and intricacy, making it vulnerable to misconfigurations or inadequate resource allocation. Furthermore, the device's ability to handle a high volume of packets and the resulting CPU and buffer utilization can lead to congestion events if not properly managed. The Alarm Code ALARM_6827, indicating a warning severity, signifies a potential issue with QoS policy enforcement or resource utilization, necessitating prompt attention from the Network Operations team to prevent service degradation.

Pre-requisite Checks

- Check: Verify redundant link status at Koramangala Datacenter

- Check: Confirm dispatch of remote hands to Koramangala Datacenter for physical access

- Check: Gather historical performance data from Node KOR-PE-01 to identify potential congestion patterns

Potential Root Causes

Cause	Probability
Misconfigured QoS policy	High
Insufficient buffer allocation for interface	Medium
High CPU utilization due to excessive packet processing	Low

Diagnostic Commands & Expected Outputs

```
Command: show mpls qos policy-map, Expected Output: Policy-map configuration and queueing information, Description: This command displays the configured policy-maps and their corresponding queueing information, aiding in the identification of potential misconfigurations or inadequate resource allocation.

Command: show interface | include Drops, Expected Output: Interface drops and error statistics, Description: This command provides detailed interface drop statistics, indicating potential tail drop events or congestion issues.
```

```
Command: show queueing interface , Expected Output: Queueing information for interface, Description: This command displays queueing information for the specified interface, including queue lengths and utilization rates, facilitating the identification of potential QoS queue starvation issues.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: show mpls qos policy-map, Description: Carefully examine the policy-map configuration and queueing information to ensure proper QoS policy enforcement. Make necessary corrections to prevent misconfigurations or inadequate resource allocation.

Step 2: Step: Adjust buffer allocation for interface, Command: interface , Description: Increase buffer allocation for the specified interface to prevent excessive queueing and potential tail drop events.

Step 3: Step: Implement CPU-intensive tasks during off-peak hours, Command: none, Description: Schedule CPU-intensive tasks, such as software upgrades or backups, during off-peak hours to minimize the impact on CPU utilization and prevent potential congestion events.

Post-Check Verification

```
Command: show mpls qos policy-map, Expected Output: Corrected policy-map configuration and queueing information, Description: Verify that the QoS policy configuration has been corrected and queueing information reflects proper resource allocation.

Command: show interface | include Drops, Expected Output: Reduced interface drops and error statistics, Description: Confirm that interface drop statistics have decreased, indicating a resolution to potential tail drop events or congestion issues.

Command: show queueing interface , Expected Output: Improved queueing information for interface, Description: Verify that queueing information for the specified interface reflects improved utilization rates and reduced queue lengths.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: config replace , Description: Restore the original QoS policy configuration from a backup file to revert any changes made during the resolution process.

- Step: Revert buffer allocation changes, Command: interface , Description: Revert buffer allocation changes made to the specified interface to restore the original configuration.